RATING METHODOLOGY

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Rating Methodology

Aerospace and Defense

This rating methodology replaces the *Aerospace and Defense* methodology published on 9 December 2024. We have added clarifying text to describe our approach to assessing the Revenue and Earnings Stability subfactor within the Business Profile factor. We have also made editorial changes throughout the methodology to enhance readability. These updates do not change our methodological approach.

Scope

This rating methodology applies to companies globally that are primarily engaged in the development, production, assembly, sale and distribution of parts and services in the aerospace and defense industry. Companies that we rate using this methodology may produce large commercial airplanes, regional jets and business jets, or supply aircraft systems or components such as engines or flight controls to commercial customers.

This methodology also applies to companies that may provide military products such as aircraft, submarines and weapons primarily to governments or to other defense contractors. Companies that we rate using this methodology may also provide maintenance, repair and overhaul, or other services, to both commercial aerospace and defense end markets.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

We rate companies that provide products or services to aerospace and defense end markets, but for which this business does not represent the preponderance of the company's business risk, using other methodologies, such as our methodologies for manufacturing companies and business and consumer services companies.[1]

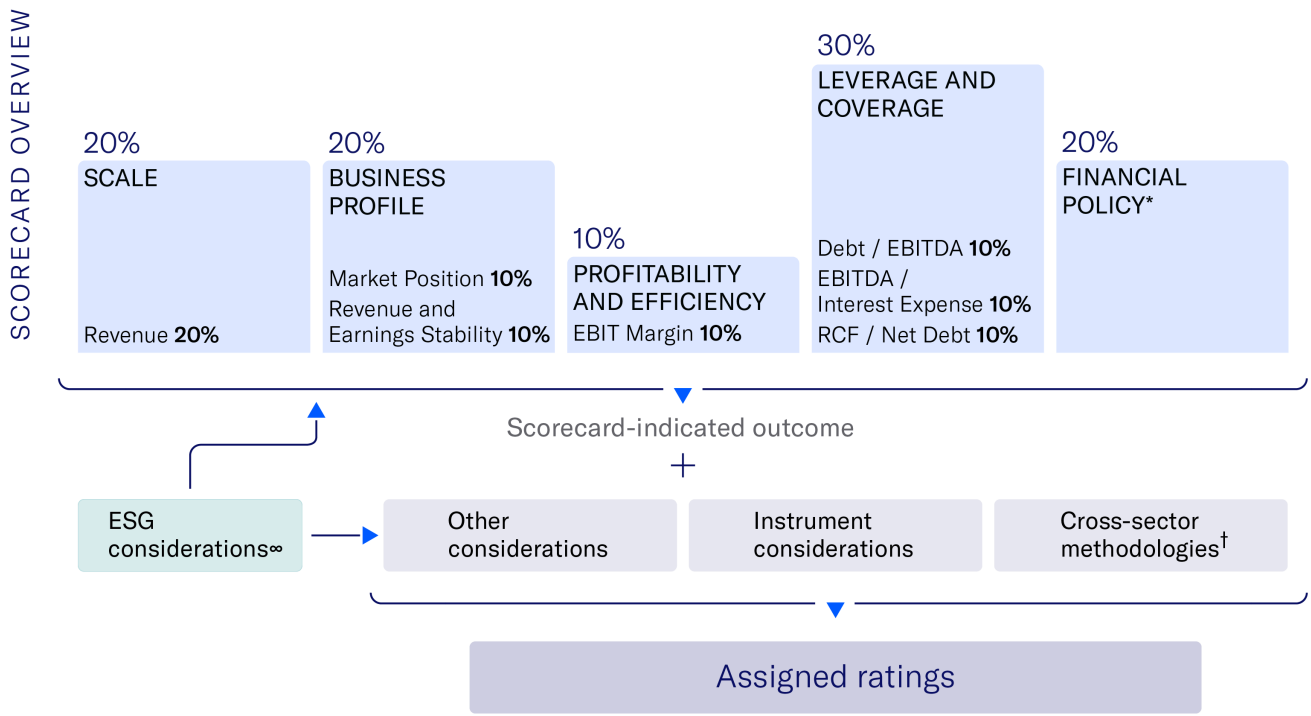
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the aerospace and defense rating approach



* This factor has no subfactors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Aerospace and defense scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion)[1]	20%	≥ $60	$30 - $60	$15 - $30	$8 - $15	$3 - $8	$1 - $3	$0.25 - $1	< $0.25
Factor: Business Profile (20%)									
Market Position	10%	Sole provider or leading market position in essentially all segments; prime position on essentially all contracts; essentially no alternatives to products or services; essentially all programs are highly strategically important to customer(s); and extremely high barriers to entry in essentially all segments.	Leading market position in essentially all segments; prime position on most contracts; few alternatives to products or services; most programs are strategically important to customer(s), and very high barriers to entry in essentially all segments.	Leading or No. 2 market position in most segments; prime or Tier 1 position on most contracts; few alternatives to products or services; most programs are strategically important to customer(s); very high barriers to entry in most segments.	Leading or No. 2 market position in most core segments; Tier 1 position on most contracts; few alternatives to products or services; many programs are important to customer(s); high barriers to entry in most segments.	Strong market position in most core segments; Tier 1 or Tier 2 positions on most contracts; alternatives for some products and services; several programs are important to customer(s) and some are being phased out; high barriers to entry in most core segments.	Solid market position in some core segments; mostly Tier 2 positions on contracts; alternatives for most products and services; a few programs are important to customer(s) and some are being phased out; moderately high barriers to entry in some segments.	Weak market position in most segments; low-tier position on most contracts; most products and services are easily replicable; most programs lack importance to customer(s) or are being phased out; or low barriers to entry in most segments.	Low-tier position on essentially all contracts; essentially all products and services are easily replicable; essentially all programs lack importance to customer(s) or are being phased out; or most segments have essentially no barriers to entry.
Revenue and Earnings Stability	10%	Extremely high revenue and earnings stability, supported by exceptional, multi-year backlog; essentially all revenue derived from extremely stable end markets with long product cycles; essentially no exposure to budget cuts, deferrals, cancellations or cyclicality; and exceptional track record of execution.	Very high revenue and earnings stability, supported by extremely strong, multi-year backlog; revenue derived mostly from extremely stable end markets with long product cycles; essentially no exposure to budget cuts, deferrals, cancellations or cyclicality; and extremely strong track record of execution.	High revenue and earnings stability, supported by very strong, multi-year backlog; revenue derived mostly from highly stable end markets with long product cycles; very low exposure to budget cuts, deferrals, cancellations or cyclicality; very strong track record of execution.	High revenue and earnings stability in most segments, supported by strong, multi-year backlog; revenue derived mostly from stable end markets with long product cycles; low exposure to budget cuts, deferrals, cancellations or cyclicality; strong track record of execution.	High revenue and earnings stability in some segments, supported by fairly strong backlog; some revenue derived from moderately stable end markets with long product cycles; may have some exposure to budget cuts, deferrals, cancellations or cyclicality; fairly strong track record of execution.	Moderate volatility of revenue and earnings in most segments; moderate backlog; limited revenue derived from stable end markets; some exposure to budget cuts, deferrals, cancellations or cyclicality; moderate track record of execution.	High revenue and earnings volatility; limited backlog; very limited revenue from stable end markets; high exposure to budget cuts, deferrals, cancellations or cyclicality; or weak track record of execution.	Extremely volatile revenue and earnings; limited or non-existent backlog; essentially no revenue from stable end markets; very high exposure to budget cuts, deferrals, cancellations or cyclicality; or extremely weak track record of execution.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Profitability and Efficiency (10%)									
EBIT Margin[2]	10%	≥ 25%	20% - 25%	15% - 20%	10% - 15%	7% - 10%	4% - 7%	1% - 4%	< 1%
Factor: Leverage and Coverage (30%)									
Debt / EBITDA[3]	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 8x	> 8x
EBITDA / Interest Expense[4]	10%	≥ 20x	15x - 20x	10x - 15x	5x - 10x	2.5x - 5x	1x - 2.5x	0.5x - 1x	< 0.5x
RCF / Net Debt[5]	10%	≥ 70%	50% - 70%	35% - 50%	20% - 35%	13% - 20%	5% - 13%	0% - 5%	< 0%
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 40%. A value of 40% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 30x. A value of 30x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 19.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (20% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Market Position

Market position is an important indicator of a company's resilience to downturns and ability to maintain a strong business profile over time. Core aspects of an aerospace and defense company's market position include the strength of its position on contracts, the importance of its programs to customers, and barriers to entry in its business segments.

A leading market position may indicate that a company's investments in research and development are translating into competitive advantages and that its products and services are meaningfully differentiated from competitors.

An aerospace and defense provider with a prime or Tier 1 position on a contract is typically difficult to replace and has staying power, whereas a lower-tier company that sells undifferentiated components is generally more easily replaced by the customer and may experience greater pricing pressure. An aerospace and defense company with prime position typically supplies products or services directly to the end customer. For example, an airplane manufacturer that sells planes directly to an airline or provides services directly to a government has a prime position. A company that sells products to the prime airplane manufacturer may have a Tier 1 position.

Suppliers to the Tier 1 provider are generally considered to be lower-tier companies, typically indicating greater distance from the end customer.

An aerospace and defense company generally provides products and services to an end customer as part of the customer's overall spending program. An aerospace and defense company that derives sales from programs that are strategically important to the customer is typically more likely to have sustained revenue from that customer over time, whereas sales from programs that are less critical or ones that are being phased out typically indicate lower future revenue opportunity.

Barriers to entry provide important indications of the likelihood of competition developing among aerospace and defense companies and of customers switching providers or remaining loyal. Companies that offer products and services with a higher level of intellectual property and sophisticated engineering, for example, typically benefit from greater pricing power.

Gaining regulatory approval for products or special clearance for employees is another barrier to entry because it is generally costly and time consuming and creates inherent limitations on the number of providers of these products or employees. Companies that offer products and services that have cleared regulatory hurdles are therefore typically at an advantage. In some defense contracts, the customer (e.g., a government) shares highly classified information and may be less likely to switch providers. Barriers to entry may be particularly important in cases where the company pays an upfront program participation fee for a contract and anticipates future revenue over the contract term to recoup this fee and achieve profitability.

Revenue and Earnings Stability

Revenue and earnings stability helps a company better withstand short-term economic cycles or industry disruptions. Core aspects of a company's revenue and earnings stability include its backlog, the stability of its end markets and its track record of execution.

Backlog, which represents committed orders from customers, indicative orders, or expected revenue based on long-term programs, typically indicates a minimum level of revenue a company is likely to generate. A company with a strong, multiyear backlog can typically expect more revenue stability than one with a smaller backlog that may be subject to greater uncertainty. Backlog may be a less important consideration for suppliers with products that are less capital-intensive and require less time to develop.

Exposure to multiple end markets across and within the broad categories of aerospace and defense typically provides a company stability, because the company derives revenue streams from end markets and products that are subject to different economic and industry cycles and trends. For example, an aeroengines manufacturer that supplies both the commercial aviation and defense sectors can benefit from steadier revenue and earnings than can a company solely focused on commercial markets. Companies in the defense business are typically more insulated from downturns because defense contracts follow cycles linked to government budgets rather than economic cycles. Additionally, product and platform diversity generally leads to revenue and earnings stability. Having various products spanning multiple platforms at different points in their life cycle helps mitigate a company's dependency on certain end markets and vulnerabilities to supply chain disruptions.

Products with longer life cycles can also help an aerospace and defense company's revenue and earnings stability. For example, large airplanes used for long commercial flights are typically flown for more than 10 years and also require a lengthy manufacturing process; therefore, orders are placed well in advance.

Exposure to business in different regions may also help a company mitigate economic or other weakness in one region with more favorable trends in another. For example, governments typically commit to defense budgets over multiple years, which means that a company that sells to multiple regions may be able to offset weakening revenue from one country's declining defense budget with business in a country that is increasing defense spending.

Because aerospace and defense contracts often involve complex products and services, a company's track record of delivery of such products and services is generally an important indication of the company's ability to convert its backlog into a consistent revenue stream over time. A company's track record of execution may also provide important indications of its ability to adapt to a customer's changing requirements and to manage its cost structure. A company with a poor track record of execution is typically less likely to maintain its current contracts or to win new orders from existing customers.

How we assess it for the scorecard

We score this factor based on two qualitative subfactors: Market Position; and Revenue and Earnings Stability.

We generally do not expect a given company to match each attribute listed for a given scoring category for each subfactor. We typically assign a subfactor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on a subfactor score.

Market Position

In our assessment of a company's market position, we consider whether a company has a prime, Tier 1, or lower-tier position across its key contracts, as well as whether competitors offer similar products and services. We may consider market share, which we generally estimate based on information provided by third parties and a company's financial disclosures, and we may also use nonpublic information. Companies with mostly lower-tier positions on key contracts typically score lower for this subfactor, although a company that is the sole source of a key product to another provider that has a prime or Tier 1 position may have a strong market position. A company that primarily offers commodity products and services that can be easily substituted typically scores lower for this subfactor than one that offers more sophisticated products and services.

Our assessment of the importance of an aerospace and defense company's program to customers involves qualitative considerations of key contracts and customers. We consider whether a program is critical to the end customer's long-term strategy or whether the customer is phasing out the product or service. We may consider the customer's public commitments to particular programs, such as a government's statements about its budget priorities. For example, an aerospace and defense company that manufactures energy-efficient airplanes is likely to be more competitive because airlines typically replace older planes with more energy-efficient ones over time. For a defense contractor, a program targeted at modernizing a government's information technology equipment or supporting a critical national security program is more likely to be a strategic priority for the customer than one related to older equipment.

Exhibit 3 summarizes common barriers to entry for aerospace and defense companies. We also typically assess whether the company is investing to maintain its unique offering, because barriers may erode without continued investment.

Exhibit 3

Typical barriers to entry

Barrier	Description
Investment Intensity	Substantial, ongoing investment in physical assets for complex sub-assemblies, and the intellectual capacity needed to create and maintain systems or electronics/surveillance operations. Could include the need for personnel with advanced degrees, unique skills or high clearances.
Unique Technologies	Patented, or difficult to replicate, products or services that provide a clear differentiated advantage in the market and are identifiable and recognized. These products, services or systems may be protected by law or be difficult to replicate by a competitor because of the time or cost involved.
Classified Programs	Large, sustainable and highly classified or secret programs (e.g., intelligence-gathering) where a defense customer is unlikely to switch providers based on costs alone.
Incumbency and Relevance	Important to suppliers as well as lead program managers or contractors. Over the life of a program, it is not possible to substitute products in many cases or only at a very high cost. Programs in civil aerospace or defense are highly consistent with the long-term direction of civilian or defense priorities.

Source: Moody's Ratings

Revenue and Earnings Stability

In assessing revenue and earnings stability, we consider a company's historical and expected revenue and earnings trends.

We generally estimate the backlog-to-sales ratio as a company's total reported backlog over its last 12 months of revenue. A high ratio typically indicates that a company has good visibility into its expected revenue. We typically use a company's reported backlog where available, or our own estimates based on market conditions and the characteristics of a company's business compared with other companies. Companies typically do not follow identical standards for reporting backlog, which may reduce comparability across issuers. We consider this qualitatively in our assessment.

The backlog-to-sales ratio may be less relevant in assessing the revenue and earnings stability of companies that provide shorter-cycle products than that of companies which offer a more complex suite of products with longer lead times.

We also assess the stability of end markets for a company's major contracts, including exposure to a customer's budget cuts, deferrals, cancellations or cyclicality. The defense market is typically more stable than the commercial market, and it also generally follows government budget cycles rather than economic cycles. As such, a mix of defense and commercial business may result in stronger overall revenue and earnings stability, depending upon the type of commercial business. Within the commercial business, we consider the mix of end customers, products and platforms, which may affect revenue and earnings stability. For example, sales of business jets are typically more cyclical than commercial jets, which tend to benefit from a more predictable, multiyear cycle. We may also consider geographic diversity as part of our revenue and earnings stability assessment.

We assess a company's track record of contract execution over time, and we consider its ability to manage its supply chain. When assessing existing contracts, we evaluate contractual protections that may help insulate a company from order cancellations or cost increases, including those driven by inflation and regulatory changes. Additionally, we consider the relative mix of development-phase and in-production contracts, given the potential for execution challenges and cost overruns as contracts ramp up. A company that primarily offers commodity products and services is unlikely to have a strong track record of contract execution, because the lack of complexity in its products and services does not afford it with the opportunity to develop one. We may also consider a company's governance culture and practices in our assessment of contract execution.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[3]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (30% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: Debt/EBITDA, EBITDA/Interest Expense and RCF/Net Debt.[4]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBITDA / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to interest expense.

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Aerospace and defense companies may face environmental risks related to air, soil and water pollution created during manufacturing, as well as the resulting costs of cleanup. Some companies sell to airlines, an end-market sector that faces pressure to reduce its carbon emissions. Airline requirements to reduce carbon emissions drive demand for more fuel-efficient aircraft, which may create advantages for aerospace and defense companies that can deliver equipment with higher fuel efficiency and disadvantages for those that cannot keep up with requirements. In most developed markets, clear regulation and compliance by companies may reduce the risk of unexpected costs. Companies with operations in countries with less rigorous standards may face lower regulatory costs but possibly

higher cleanup risks. The impact of these risks typically varies based on a company's business mix. For example, companies that sell primarily high-value-added products and services can generally pass on to customers cost increases related to regulations, whereas companies that sell more commoditized products typically have to absorb cost increases.

Social considerations relevant to aerospace and defense companies relate primarily to labor and product safety. A shrinking pool of more highly skilled workers, such as engineers, may pose a risk for some companies as they seek to keep pace with technological change. Workers in certain hazardous industries may face potential health and safety risks. Product recalls or redesigns may affect companies' supply chains. Most aerospace and defense companies sell to other businesses rather than to consumers, but the airline sector is a key end market for companies that produce airplanes or supply related systems, components or services. Aerospace and defense companies with a concentration of sales to airlines may have exposure to changing consumer preferences if increasing awareness of the potential environmental damage from emissions reduces demand for air travel.

Governance considerations relevant to aerospace and defense companies may include management's oversight over contracts.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's Market Position and Revenue and Earnings Stability subfactors. However, we may assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or

under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we often consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

For example, free cash flow is not always an important differentiator of credit profiles. Strong companies with excellent investment opportunities may demonstrate multiyear periods of negative free cash flow while retaining solid access to capital and credit, because these investments will yield stable cash flow in future years. Weaker companies with limited access to credit may have positive free cash flow for a period of time because they have curtailed the investments necessary to maintain their assets and future cash-generating prospects. However, in some cases, free cash flow can be an important driver of the future liquidity profile of a company.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile.

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Seasonality

Seasonality is an important driver of customer demand and can cause swings in cash balances and working capital positions for companies in this sector. Higher volatility creates less room for errors in meeting customer demand or operational execution.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 4

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 5

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 6
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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